

PinitHUB — Project Build Brief

Unified platform for image & document protection, forensic verification, marketplace, and platform administration

PREPARED BY

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SCOPE

Landing, CMS Admin,
Pixel Protection, Master
Admin

DATE

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1. Overview

PinitHUB is a forensic-grade platform for protecting images and documents, verifying their authenticity, and giving creators a marketplace to license their work. Four major pieces of the system have been built out: the public marketing site, an internal CMS to run that site, the pixel-level protection engine that is the core of the product, and a Master Admin console that gives the platform team full operational visibility across users, assets, investigations, and revenue.

2. Landing Page (PINITHUB Marketing Site)

A complete public marketing site for the platform, redesigned around an enterprise SaaS narrative with clear calls-to-action into the live product.

- Built on Next.js 15 (App Router), Tailwind CSS v4, Framer Motion for animation, and three.js for 3D visuals
- Enterprise-narrative redesign: positions PINITHUB as the parent brand and PinIT Hub as the live product
- Product demo video embedded on the hero section with a dedicated "Watch demo" flow
- Split into two independent Vercel deployments — a public surface and an admin surface — so the CMS login is never reachable from the public marketing URL

3. Admin / CMS for the Landing Page

A dedicated internal admin app for editing the marketing site's content without a code deploy, isolated from the public site for security.

- Authenticated admin login, separate from the public site's routing entirely (returns 404 on the public deploy)
- Content management for page sections and structured collections
- Inbox for demo requests and newsletter sign-ups captured from the public site
- Admin user management and an audit log of admin actions
- Shares one Postgres database (Prisma) with the public site; content edits go live via on-demand revalidation

4. Pixel-Level Protection — Images & Documents

The core forensic engine: every protected file gets a cryptographic identity down to the pixel level, so tampering can be proven and localized, not just detected.

- Hierarchical spatial authentication — files are broken into a grid of blocks, each independently keyed and authenticated, so a tampered region can be pinpointed rather than just flagging "the file changed"
- Pixel-level keyed authentication scheme (HKCA) already iterated through two hardened versions (v1 → v1.1, with a stronger binary crypto serialization) without breaking older protected files
- Fragment / splice detection — separately catches a small piece of a protected original (e.g. a cropped face) pasted into an unrelated image, which whole-image comparison would miss
- Document protection extended to PDFs: pages are rasterized and run through the same pixel-level protection pipeline as standalone images, with configurable render quality and a page cap for cost control
- Full investigation tooling on top: tamper localization, forensic scanning, and a unified investigation service that ties matches back to a case an investigator can own and close

5. Master Admin Dashboard — Whole-Platform Operations

A standalone admin console (its own app, independent of the main product) giving the platform team a single place to operate every part of PinitHUB.

- **Command Center** — live KPIs (users, organizations, assets, DNA-protected files, revenue), activity trends, and critical alerts on one screen
- **Users & Organizations** — full directory with drill-down into any user or org
- **Assets & DNA** — every protected file and its fingerprint history
- **Intelligence (Sentinel)** — investigations, tamper findings, and forensic detections, each resolvable into an owned case
- **Marketplace, Billing & Credits** — subscriptions, revenue, and usage against plan limits
- **Network Intelligence & Web Monitoring** — organizational reach and crawler-based leak detection
- **Identity & Verification, Support & Disputes, Audit Logs, Reports** — the remaining day-to-day operational surfaces, all in one console
- Backed by capability-scoped role-based access control and an admin audit trail, so every admin action is permissioned and logged
- Rebuilt this week as its own dedicated app with a full visual pass — corrected a layout bug that was leaving half the screen blank, and reworked the type scale for legibility on the ops team's displays

6. Status Snapshot

Area	Status
Landing Page (public marketing site)	Live
Landing Page Admin / CMS	Live
Pixel-Level Protection — Images	Live
Pixel-Level Protection — Documents (PDF)	Live
Master Admin Dashboard	In Progress — core built, visual pass underway

7. Technology Stack

Layer	Technologies
Frontend	React, TypeScript, Next.js 15, Vite, Tailwind CSS, Framer Motion, three.js
Backend	Node.js, Express, TypeScript, FastAPI (Python AI sidecar)
Database	PostgreSQL (Supabase), Prisma ORM
Storage	Supabase Storage (encrypted vault)
Security	Keyed pixel-level authentication (HKCA), AES-256-GCM encryption, JWT auth, RBAC
Deployment	Vercel (frontend/CMS), Render (backend + AI service)